

Generation Interconnection System Impact Study Report

For Queue Project Q19-1179

GI Dairy Biogas 138 kV

March 15, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1179. The Interconnection Customer proposes a 21.2 MW solar photovoltaic facility interconnecting at the GI Dairy Biogas 138 kV substation in the Pennsylvania area. The request was received on April 01, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 457 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 16 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$90,594,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: GI Dairy Biogas / Hills Chapel 138 KV L
Contingency: Suffolkaire / United Power 138 KV
Loading: 141.0% (Rating: 198 MVA)
DFAX: 70.2%
Pre-project loading: 133.5%
2. Facility: Alta / W Dallas Tap 138 KV
Contingency: Burns / Hills Chapel 345 KV
Loading: 136.4% (Rating: 238 MVA)
DFAX: 31.8%
Pre-project loading: 133.6%
3. Facility: Livingston Westside Substation / Fern Hill 138 KV
Contingency: Burns / Hills Chapel 345 KV L
Loading: 130.2% (Rating: 261 MVA)
DFAX: 31.8%
Pre-project loading: 127.6%
4. Facility: Livingston Westside / United Power 138 KV
Contingency: Burns / Hills Chapel 345 KV
Loading: 128.1% (Rating: 270 MVA)
DFAX: 31.8%
Pre-project loading: 125.6%
5. Facility: Hills Chapel 345/161 kV XFMR
Contingency: Wallenpaupack 500-345 kV Autotransformer
Loading: 125.5% (Rating: 311 MVA)
DFAX: 35.6%
Pre-project loading: 123.1%
6. Facility: Hills Chapel 161/138 kV Auto
Contingency: United Power / Livingston Westside 138 KV
Loading: 125.5% (Rating: 164 MVA)

DFAX: 64.9%

Pre-project loading: 117.1%

7. Facility: Fern Hill / W Dallas Tap Substation 138 KV

Contingency: Burns / Hills Chapel 345 KV line

Loading: 111.4% (Rating: 295 MVA)

DFAX: 31.8%

Pre-project loading: 109.1%

8. Facility: Burns 345/138 kV TX

Contingency: Coronado / Burns 345 KV

Loading: 109.2% (Rating: 478 MVA)

DFAX: 6.5%

Pre-project loading: 108.9%

9. Facility: United Power / Bttas Rd 138 KV

Contingency: Burns / Hills Chapel 345 KV line

Loading: 108.2% (Rating: 190 MVA)

DFAX: 17.6%

Pre-project loading: 106.3%

10. Facility: Oak Knoll / Baker Riv Substation 138 KV

Contingency: Coronado Substation / Burns 345 KV

Loading: 105.7% (Rating: 484 MVA)

DFAX: 6.5%

Pre-project loading: 105.4%

11. Facility: Wilmington Station C / Hills Chapel 138 KV

Contingency: Hills Chapel / Burns 345 KV

Loading: 105.2% (Rating: 225 MVA)

DFAX: 17.6%

Pre-project loading: 103.6%

12. Facility: Bettas Rd / Huron 138 KV

Contingency: Burns / Hills Chapel 345 KV

Loading: 105.2% (Rating: 202 MVA)

DFAX: 17.6%

Pre-project loading: 103.3%

13. Facility: Baker Riv / Burns 138 KV

Contingency: Burns / Coronado 345 KV

Loading: 104.5% (Rating: 500 MVA)

DFAX: 6.5%

Pre-project loading: 104.2%

14. Facility: Lakeshore Mine / Alta 138 KV

Contingency: Dearborn Industrial Generation Substation / Alta 138 KV

Loading: 104.0% (Rating: 168 MVA)

DFAX: 27.0%

Pre-project loading: 100.6%

15. Facility: Hamilton Dome / Oak noll 138 KV

Contingency: Burns / Coronado 345 KV

Loading: 103.4% (Rating: 469 MVA)

DFAX: 6.5%

Pre-project loading: 103.2%

16. Facility: Wilmington Sta C / Huron 138 KV

Contingency: Burns / Hills Chapel 345 KV

Loading: 103.2% (Rating: 218 MVA)

DFAX: 17.6%

Pre-project loading: 101.5%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
GI Dairy Biogas / Hills Chapel 138 KV L	Rebuild line with higher-capacity conductor	\$12,324,000	100% (\$12,324,000)
Alta Substation / West Dallas Tap 138 KV	Rebuild line with higher-capacity conductor	\$9,338,000	100% (\$9,338,000)
Livingston Westside Substation / Fern Hill 138 KV	Rebuild line with higher-capacity conductor	\$5,862,000	100% (\$5,862,000)
Livingston Westside / United Power 138 KV	Rebuild line with higher-capacity conductor	\$3,520,000	100% (\$3,520,000)
Hills Chapel 345 kV/161 kV TX	Replace transformer with higher-rated unit	\$28,386,000	31% (\$8,819,000)
Hills Chapel 161 kV/138 kV TX	Replace transformer with higher-rated unit	\$7,144,000	100% (\$7,144,000)
Fern Hill / W Dallas Tap 138 KV	Rebuild line with higher-capacity conductor	\$2,379,000	100% (\$2,379,000)
Burns 345 kV/138 kV Transformer	Replace transformer with higher-rated unit	\$15,531,000	100% (\$15,531,000)
United Power / Bettas Rd Substation 138 KV	Reconductor line	\$3,882,000	100% (\$3,882,000)
Baker River / Oak Knoll 138 KV L	Reconductor line	\$4,300,000	100% (\$4,300,000)
Wilmington Sta C / Hills Chapel 138 KV line	Reconductor line	\$3,380,000	100% (\$3,380,000)
Bettas Road / Huron 138 KV	Reconductor line	\$3,537,000	100% (\$3,537,000)
Baker Riv / Burns 138 KV line	Reconductor line	\$970,000	100% (\$970,000)
Lakeshore Mine / Alta 138 KV	Reconductor line	\$5,762,000	100% (\$5,762,000)
Hamilton Dome / Oak noll 138 KV	Reconductor line	\$908,000	100% (\$908,000)
Wilmington Sta C / Huron 138 KV	Reconductor line	\$2,938,000	100% (\$2,938,000)
	Total allocated to project		\$90,594,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.